IRRIGATION WATER MANAGEMENT · NOTE IM03

Sprinkler Irrigation — Components, Distribution Pattern, Uniformity and Design

Course topics: IDE7.5 L2, IDE7.5 L3, IDE-505 T1, IDE-505 T6, IDE-501 T10, CAE1.3 L12

The method

Water is sprayed into the air through nozzles at a rate below the soil's infiltration rate, with pressure from pumping.

- **Suits** almost all crops (except rice); sandy (high-infiltration) and shallow soils; steep or irregular topography. **Unsuited** to very fine-textured heavy clays with infiltration below 4 mm/hr.
- **Advantages:** little land levelling; no permanent field channels; controlled application rate; less deep percolation and runoff; fertilisers, herbicides and fungicides can be applied through it; protects against frost and high temperature; less labour than surface irrigation.
- **Limitations:** high first cost; needs clean water; strong wind disturbs distribution; high pumping cost for the pressure.

Components and types

Pump; main, sub-main and lateral pipelines; sprinkler heads.

Type	Features
Perforated-pipe system	Holes in the laterals; low pressure 0.5–2.5 kg/cm²; wets a strip 6–15 m wide; for lawns, gardens and small vegetable fields; needs filtered water to avoid clogging
Rotating-head system	Small nozzles on risers, turned by the water's thrust (e.g. twin-nozzle heads); on the ground or above the crop

Layouts: portable, semi-permanent, solid-set, and permanent (buried mains, sub-mains and laterals).

Fertiliser through sprinklers (fertigation)

- Inject soluble fertiliser with a venturi device so irrigation and fertilising happen together, placing the nutrient at the desired depth and saving labour.
- Use a solution of 1 kg fertiliser in 30 kg water; complete injection in about 30 minutes; run the system 20–30 minutes more to flush the lines.

$$W_f = D_s \times D_l \times N_s \times W_F \,/\, 10{,}000$$

W_f = fertiliser per setting (kg); D_s = distance between sprinklers (m); D_l = distance between laterals (m); N_s = number of sprinklers; W_F = recommended dose (kg/ha).

> Sprinklers 12 m × 18 m, 10 sprinklers per setting, dose 60 kg/ha: $W_f = 12 \times 18 \times 10 \times 60 / 10{,}000 = 12.96$ kg per setting.

Distribution pattern

- The sprinkler head is the most important component; its performance at the right pressure and under the local wind decides suitability and design (along with crop water need, water-holding

capacity and infiltration).

- **Pressure:** too low — uneven distribution; satisfactory — a proper conical pattern; too high — water concentrated near the sprinkler.
- **Wind** distorts the pattern, more so at higher speeds — allow for it in spacing.
- Under proper pressure and calm conditions the pattern is a cone: depth falls with distance from the sprinkler, and infiltration follows it since the rate is below the infiltration rate. One sprinkler cannot water uniformly — overlap the wetted circles to get a nearly uniform total depth.

Testing a sprinkler's pattern

1. Run one sprinkler on a riser of the desired height at the desired pressure under the given climate.
2. Collect water in uniform catch cans on a square grid — about 2 m apart for a sprinkler spacing of about 10 m, about 3 m for larger spacings — covering the whole spread (above the crop in a cropped field).
3. Run until the smallest catch is at least 1 cm but no can overflows (one-hour runs are convenient); then measure each catch with a graduated cylinder.
4. Measure the sprinkler's pressure (gauge on a pitot tube) and discharge (timed container) before and after; evaporation loss = discharge - total catch.
5. Record wind direction and speed, temperature, humidity and any crop damage from drop impact, jet trajectory or too short a riser.

$$\text{Christiansen uniformity coefficient: } Cu = 100\ (1 - \Sigma X\ /\ (m\ n))$$

m = average application (mm/hr); n = number of observations; X = deviation of each observation from the average (so mn = total of observations). 100% is perfectly uniform; ≥ 85% is acceptable. (The textbook example gives Cu ≈ 90.4%.)

> Catches 10, 12, 9, 11, 13, 9 mm: mean m = 10.67, deviations 0.67, 1.33, 1.67, 0.33, 2.33, 1.67 (ΣX = 8.0); Cu = 100 (1 - 8.0/64) = 87.5% — acceptable.

Designing a sprinkler system

1. **Inventory:** map the size, shape and layout of the field, roads, buildings, trees, irrigated area and water-source and pump locations; check the water source for quantity, dependability (seasonal and annual crop demand) and quality — non-corrosive and free of suspended matter.
2. **Climate:** design for the daily peak consumptive use; more overlap as wind rises (table below). Allow for spray evaporation and crop interception.
3. **Soil:** keep the application rate within the infiltration rate; set the depth by the available moisture and the root-zone depletion pattern.
4. **Power:** electric pumps are cheaper and handier for stationary systems; portable diesel pumps for portable ones.
5. **Depth and rotation cycle:** depth from the root-zone moisture capacity and extraction pattern; the rotation cycle is the allowable time between irrigations at peak use (e.g. ≈ 7.5 days → say 8 days).
6. **Pump capacity** from the area, depth, interval, hours of operation and efficiency (formula below).
7. **Application rate:** match it to soil, crop cover and slope to avoid puddling and runoff (measure the infiltration with an infiltrometer).
8. **Spacing, discharge and pressure:** run the main along the dominant slope and the laterals across it, roughly perpendicular to the prevailing wind; choose the widest spacing that gives an acceptable Cu

— typically 18 × 12 m, 18 × 9 m, 15 × 12 m or 15 × 9 m.

9. **Riser height:** just above the crop; low-pressure under-tree sprinklers save much energy.

Wind speed	Maximum sprinkler spacing (% of wetted diameter)
No wind	65%
0–6.5 km/h	60%
6.5–13 km/h	50%
above 13 km/h	30%

Condition	Spray evaporation + interception loss
Bare soil, low wind	4–10%
Growing crop, low wind	3.5–5%
Bare soil, high wind	6–10%
Growing crop, high wind	7–16%

$$\text{System capacity: } Q = 27.8\ A\ D\ /\ (F\ H\ E)$$

Q = pump discharge (L/s); A = area (ha); D = depth of application (cm); F = irrigation interval (days); H = operating hours per day; E = field efficiency (fraction). (With D in mm the coefficient is 2.78.)

Corrected coefficient. The notes print Q = 278 A D/(F H E). Check the units: A ha × D cm = 100 A D m³ = 100,000 A D litres; spread over F × H × 3600 seconds this gives 27.8 A D/(F H E) L/s. "278" is ten times too large.

8 ha, depth 5 cm, interval 8 days, 20 h/day, efficiency 0.75: Q = 27.8 × 8 × 5 / (8 × 20 × 0.75) = 1112 / 120 ≈ 9.3 L/s.

$$\text{Sprinkler discharge: } q = I\ S_1\ S_2\ /\ 1000$$

q = discharge (m³/hr); I = application rate (mm/hr); S_1 = spacing along the lateral (m); S_2 = spacing between lateral positions (m).

$$\text{Application rate (cm/hr)} = \text{discharge (L/s)} \times 360\ /\ (\text{sprinkler spacing} \times \text{lateral move})\ [= 3600\ q\ /\ (S_1 S_2) \text{ in mm/hr}]$$

Sources: ICAR eCourse 'Irrigation Engineering' (IIT Kharagpur) — surface, sprinkler and drip irrigation lessons; ICAR eCourse 'Micro Irrigation Systems Design' (AgriMoon). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Sprinklers are not suitable where infiltration rate is below about

(a) 4 mm/h (b) 25 mm/h
(c) 50 mm/h (d) 100 mm/h

Q2. An acceptable Christiansen uniformity coefficient is at least

(a) 50% (b) 70%
(c) 85% (d) 99%

Q3. With winds of 6.5–13 km/h, sprinkler spacing should not exceed ___ of the wetted diameter

(a) 30% (b) 50%
(c) 60% (d) 65%

Q4. Laterals should be laid

(a) along the slope, parallel to wind (b) across the slope, roughly perpendicular to wind
(c) in circles (d) anywhere

Q5. Operating pressure of a perforated-pipe sprinkler system is about

(a) 0.05–0.25 kg/cm² (b) 0.5–2.5 kg/cm²
(c) 5–10 kg/cm² (d) 25 kg/cm²

Q6. For A ha, D cm, F days, H h/day and efficiency E, pump capacity in L/s is

(a) 2.78 AD/(FHE) (b) 27.8 AD/(FHE)
(c) 278 AD/(FHE) (d) AD/(FHE)

Answers

Q1 — (a) 4 mm/h
Q2 — (c) 85%
Q3 — (b) 50%
Q4 — (b) across the slope, roughly perpendicular to wind
Q5 — (b) 0.5–2.5 kg/cm²
Q6 — (b) 27.8 AD/(FHE) — *Unit check; 2.78 if D is in mm.*